

- 6.1
1) Find missing coordinate of P, using the fact that P lies on unit circle in given quadrant.

Coordinate: $P\left(\frac{3}{4}, \quad\right)$

Quadrant: I

$$\sqrt{\left(\frac{3}{4}\right)^2 + y^2} = 1$$

$$\frac{9}{16} + y^2 = 1$$

$$y^2 = \frac{7}{16}$$

$$y = \boxed{\frac{\sqrt{7}}{4}}$$

- 2) Point P is on unit circle. Find $P(x, y)$ from given info.
y-coordinate of P is $-\frac{4}{5}$, x-coordinate is positive

$$\sqrt{x^2 + \left(-\frac{4}{5}\right)^2} = 1$$

$$x^2 + \frac{16}{25} = 1$$

$$x^2 = \frac{9}{25}$$

$$x = \frac{3}{5}$$

$$P(x, y) = \boxed{\left(\frac{3}{5}, -\frac{4}{5}\right)}$$

- 3) Point P is on unit circle. Find $P(x, y)$ from given info.
x-coordinate of P is $-\frac{\sqrt{11}}{4}$, and $\underbrace{P \text{ lies below x-axis}}_{\text{y-coordinate is negative}}$

$$\sqrt{\left(-\frac{\sqrt{11}}{4}\right)^2 + y^2} = 1$$

$$\left(-\frac{\sqrt{11}}{4}\right)^2 + y^2 = 1$$

$$\frac{11}{16} + y^2 = 1$$

$$y^2 = \frac{5}{16}$$

$$y = \frac{\sqrt{5}}{4}$$

$$P(x, y) = \boxed{\left(-\frac{\sqrt{11}}{4}, \frac{\sqrt{5}}{4}\right)}$$

- 4) Find terminal point $P(x, y)$ on unit circle determined by given value of t .

$$t = \frac{5\pi}{2}$$

$$t = \frac{5\pi}{2} - \frac{4\pi}{2}$$

$$t = \frac{\pi}{2}$$

$$P(x, y) = \boxed{(0, 1)}$$

- 5) Find terminal point $P(x, y)$ on unit circle determined by given value of t .

$$t = \frac{3\pi}{4}$$

$$\bar{t} = \pi - \frac{3\pi}{4} \therefore \text{Q II}$$

$$\bar{t} = \frac{\pi}{4}$$

$$P(x, y) = \boxed{\left(-\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}\right)} \therefore \text{Q II}$$

6) Find terminal point $P(x, y)$ on unit circle determined by given value of t .

$$t = -\frac{2\pi}{3}$$

$$t = -\frac{2\pi}{3} + \frac{6\pi}{3}$$

$$t = \frac{4\pi}{3}$$

$$\bar{t} = \frac{4\pi}{3} - \pi \therefore \text{Q III}$$

$$\bar{t} = \frac{\pi}{3}$$

$$P(x, y) = \left(-\frac{1}{2}, -\frac{\sqrt{3}}{2}\right) \therefore \text{Q III}$$

7) Find reference number for each value of t

(a) $t = 5\pi$

$$t = 5\pi - 4\pi$$

$$t = \pi$$

$$\bar{t} = 0$$

(b) $t = -\frac{5\pi}{4}$

$$t = -\frac{5\pi}{4} + \frac{8\pi}{4}$$

$$t = \frac{3\pi}{4}$$

$$\bar{t} = \pi - \frac{3\pi}{4} \therefore \text{Q II}$$

$$\bar{t} = \frac{\pi}{4}$$

(c) $t = \frac{25\pi}{6}$

$$t = \frac{25\pi}{6} - \frac{24\pi}{6}$$

$$t = \frac{\pi}{6}$$

$$\bar{t} = \frac{\pi}{6} \therefore \text{Q I}$$

(d) $t = 4$

$$\bar{t} = 4 - \pi \therefore \text{Q III}$$

8) Consider the following.

$$t = \frac{10\pi}{3}$$

(a) Find the reference number $\bar{t}$ for value of t

$$t = \frac{10\pi}{3}$$

$$t = \frac{10\pi}{3} - \frac{6\pi}{3}$$

$$= \frac{4\pi}{3}$$

$$\bar{t} = \frac{4\pi}{3} - \pi \therefore \text{Q III}$$

$$\bar{t} = \frac{\pi}{3}$$

(b) Find terminal point determined by t

$$(x, y) = \left(-\frac{1}{2}, -\frac{\sqrt{3}}{2}\right) \therefore Q \text{ III}$$

9) Consider the following.

$$t = -\frac{23\pi}{6}$$

(a) Find the reference number $\bar{t}$ for value of t

$$t = -\frac{23\pi}{6}$$

$$t = -\frac{23\pi}{6} + \frac{24\pi}{6}$$

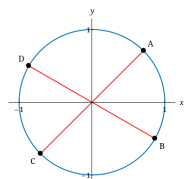
$$t = \frac{\pi}{6}$$

$$\bar{t} = \frac{\pi}{6} \therefore Q \text{ I}$$

(b) Find terminal point determined by t

$$(x, y) = \left(\frac{\sqrt{3}}{2}, \frac{1}{2}\right) \therefore Q \text{ I}$$

10) Consider the following figure.



Complete the table for each point in figure.

Point	Angle	Coordinate
A	$\frac{\pi}{4}$	$(x, y) = \left(\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}\right)$
B	$-\frac{\pi}{6}$	$(x, y) = \left(\frac{\sqrt{3}}{2}, -\frac{1}{2}\right)$
C	$-\frac{3\pi}{4}$	$(x, y) = \left(-\frac{\sqrt{2}}{2}, -\frac{\sqrt{2}}{2}\right)$
D	$\frac{5\pi}{6}$	$(x, y) = \left(-\frac{\sqrt{3}}{2}, \frac{1}{2}\right)$

Is it true in general that if the terminal point determined by t is $P(a, b)$, then the terminal point determined by $t + \pi$ is $P(-a, -b)$?

Yes

11) Show that the point is on the unit circle.

$$\left(-\frac{5}{7}, -\frac{2\sqrt{6}}{7}\right)$$

We need to show that point satisfies equation of unit circle that is $x^2 + y^2 = 1$.

$$\begin{aligned} x^2 + y^2 &= \left(-\frac{5}{7}\right)^2 + \left(-\frac{2\sqrt{6}}{7}\right)^2 \\ &= \frac{25}{49} + \frac{24}{49} \\ &= 1 \end{aligned}$$

Hence, the point is on the unit circle.

12) Consider the following point P that lies on the unit circle.

$$P\left(\frac{3}{4}, y\right)$$

Determine all possible y-coordinates for this point.

$$\sqrt{\left(\frac{3}{4}\right)^2 + y^2} = 1$$

$$\frac{9}{16} + y^2 = 1$$

$$y^2 = \frac{7}{16}$$

$$y = \frac{\sqrt{7}}{4}, y = -\frac{\sqrt{7}}{4}$$

Find missing coordinate of P, given that it lies on unit circle in Quadrant I.

positive
y-coordinate

$$P\left(\frac{3}{4}, \frac{\sqrt{7}}{4}\right)$$

13) Consider the following.

$$t = -5\pi$$

(a) Find the reference number $\bar{t}$ for value of t

$$t = -5\pi$$

$$t = -5\pi + 6\pi$$

$$t = \pi$$

$$\bar{t} = 0$$

(b) Find terminal point determined by t

$$(x, y) = (-1, 0)$$

6.2

1) Find exact value of trig function at given real number

(a) $\sin\left(\frac{4\pi}{3}\right)$

reference number:

$$\frac{4\pi}{3} - \pi \because \text{Q III}$$

$$= \frac{\pi}{3}$$

$$\sin\left(\frac{4\pi}{3}\right) = -\frac{\sqrt{3}}{2} \because \text{Q III}$$

(b)

$$\sec\left(\frac{5\pi}{6}\right) = \frac{1}{\cos\left(\frac{5\pi}{6}\right)}$$

reference number:

$$\pi - \frac{5\pi}{6} \because \text{Q II}$$

$$= \frac{\pi}{6}$$

$$\sec\left(\frac{5\pi}{6}\right)$$

$$= \frac{1}{\cos\left(\frac{\pi}{6}\right)}$$

$$= \frac{1}{\frac{\sqrt{3}}{2}}$$

$$= \frac{2}{\sqrt{3}}$$

$$= -\frac{2\sqrt{3}}{3} \because \text{Q II}$$

$$\begin{aligned}
 (c) \quad & \cot\left(-\frac{\pi}{3}\right) \\
 &= -\cot\left(\frac{\pi}{3}\right) \\
 &= -\frac{1}{\tan\left(\frac{\pi}{3}\right)} \\
 &= -\frac{1}{\sqrt{3}} \\
 &= \boxed{-\frac{\sqrt{3}}{3}}
 \end{aligned}$$

2) Find exact value of trig function at given real number

$$\begin{aligned}
 (a) \quad & \sin\left(\frac{29\pi}{2}\right) \\
 &= \sin\left(\frac{\pi}{2}\right) \\
 &= \boxed{1}
 \end{aligned}$$

$$\begin{aligned}
 (b) \quad & \cos\left(\frac{29\pi}{2}\right) \\
 &= \cos\left(\frac{\pi}{2}\right) \\
 &= \boxed{0}
 \end{aligned}$$

$$\begin{aligned}
 (c) \quad & \cot\left(\frac{29\pi}{2}\right) \\
 &= \cot\left(\frac{\pi}{2}\right) \\
 &= \boxed{0}
 \end{aligned}$$

3) Write first expression in terms of second if terminal point determined by t is in given quadrant

$\sin(t)$, $\cos(t)$; Quadrant II

$$\sin^2 t + \cos^2 t = 1$$

$$\boxed{\sin t = \sqrt{1 - \cos^2 t}} \quad \because Q II$$

4) Write first expression in terms of second if terminal point determined by t is in given quadrant

$\tan(t)$, $\sin(t)$; Quadrant IV

$$\sin^2 t + \cos^2 t = 1$$

$$\cos t = \pm \sqrt{1 - \sin^2 t}$$

$$\tan t = \frac{\sin t}{\cos t}$$

$$\tan t = \boxed{\frac{\sin t}{\sqrt{1 - \sin^2 t}}} \quad \because Q IV$$

5) Find terminal point determined by $t = \frac{8\pi}{3}$

$$t = \frac{8\pi}{3}$$

$$t = \frac{2\pi}{3}$$

reference number:

$$\pi - \frac{2\pi}{3} \quad \because Q II$$

$$= \frac{\pi}{3}$$

$$P(x, y) = \left(-\frac{1}{2}, \frac{\sqrt{3}}{2}\right) \therefore \text{Q II}$$

Find exact value of trig function at given real number t

$$(a) \sin\left(\frac{8\pi}{3}\right)$$

$$\boxed{\frac{\sqrt{3}}{2}}$$

$$(b) \csc\left(\frac{8\pi}{3}\right)$$

$$= \frac{1}{\sin\left(\frac{8\pi}{3}\right)}$$

$$= \frac{1}{\frac{\sqrt{3}}{2}}$$

$$= \boxed{\frac{2\sqrt{3}}{3}}$$

$$(c) \cot\left(\frac{8\pi}{3}\right)$$

$$= \frac{1}{\tan\left(\frac{8\pi}{3}\right)}$$

$$= \frac{1}{\frac{\frac{\sqrt{3}}{2}}{-\frac{1}{2}}}$$

$$= \boxed{-\frac{\sqrt{3}}{3}}$$

6) Find exact value of trig function at given real number

$$(a) \sin(17\pi)$$

$$= \sin(\pi)$$

$$= \boxed{0}$$

$$(b) \cos(18\pi)$$

$$= \boxed{1}$$

$$(c) \tan(19\pi)$$

$$= \tan(\pi)$$

$$= \boxed{0}$$

7) The terminal point $P(x, y)$ determined by a real number t is given.
Find $\sin(t)$, $\cos(t)$, and $\tan(t)$

$$\left(-\frac{12}{13}, -\frac{5}{13}\right)$$

$$\sin(t) = \boxed{-\frac{5}{13}}$$

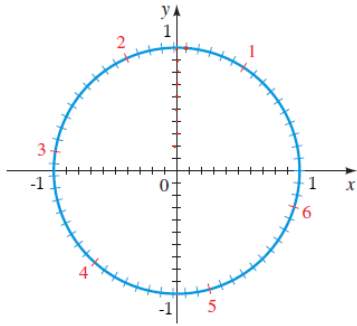
$$\cos(t) = \boxed{-\frac{12}{13}}$$

$$\tan(t) = \frac{-\frac{5}{13}}{-\frac{12}{13}}$$

$$= \boxed{\frac{5}{12}}$$

8) Find an approx value of given trig function by using following figure and a calculator

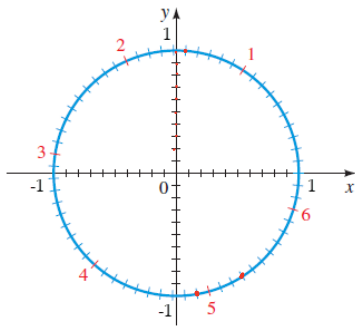
$$\sin(1.5)$$



a) figure
 ≈ 0.9

b) calculator
 ≈ 0.9975

9) Find an approx value of given trig function by using following figure and a calculator
 $\tan(-1.4)$



a) figure
 ≈ -5.8

b) calculator
 ≈ -5.7979

10) Write first expression in terms of second if terminal point determined by t is in given quadrant

$\csc(t), \cot(t)$; Q III

$$\csc^2 t = 1 + \cot^2 t$$

$$\csc t = -\sqrt{1 + \cot^2 t} \quad \because \text{Q III}$$

11) Find the values of trig functions of t from given info.

$$\csc(t) = 5, \cos(t) < 0$$

$$\sin(t) = \frac{1}{\csc(t)}$$

$$\sin(t) = \frac{1}{5}$$

$$\sin^2 t + \cos^2 t = 1$$

$$\left(\frac{1}{5}\right)^2 + \cos^2 t = 1$$

$$\cos^2 t = \frac{24}{25}$$

$$\cos t = -\frac{2\sqrt{6}}{5} \quad \because \cos(t) < 0$$

$$\tan(t) = \frac{\frac{1}{5}}{-\frac{2\sqrt{6}}{5}}$$

$$= -\frac{1}{2\sqrt{6}}$$

$$\tan(t) = -\frac{\sqrt{6}}{12}$$

$$\sec(t) = \frac{1}{-\frac{\sqrt{6}}{12}}$$

$$= -\frac{12}{\sqrt{6}}$$

$$\sec(t) = -\frac{5\sqrt{6}}{12}$$

$$\cot(t) = \frac{1}{-\frac{\sqrt{6}}{12}}$$

$$= -\frac{12}{\sqrt{6}}$$

$$\cot(t) = -2\sqrt{6}$$

12) Find the values of trig functions of t from given info.

$$\tan(t) = -3, \quad \csc(t) = 70$$

Q II, Q IV

Q I, Q II

Q II

$$\cot(t) = \frac{1}{\tan(t)}$$

$$\cot(t) = -\frac{1}{3}$$

$$\tan^2 t + 1 = \sec^2 t$$

$$(-3)^2 + 1 = \sec^2 t$$

$$\sec t = -\sqrt{10} \quad \therefore \text{Q II}$$

$$\cot^2 t + 1 = \csc^2 t$$

$$\left(-\frac{1}{3}\right)^2 + 1 = \csc^2 t$$

$$\csc t = \frac{\sqrt{10}}{3} \quad \therefore \text{Q II}$$

$$\cos(t) = \frac{1}{\sec(t)}$$

$$= \frac{1}{-\sqrt{10}}$$

$$\cos(t) = -\frac{\sqrt{10}}{10}$$

$$\sin(t) = \frac{1}{\csc(t)}$$

$$= \frac{1}{\frac{\sqrt{10}}{3}}$$

$$\sin(t) = \frac{3\sqrt{10}}{10}$$

13) Determine whether the function is even, odd, or neither

$$f(x) = x^6 \sin(x)$$

$$f(-x) = x^6 (-\sin(x))$$

$$f(-x) = -x^6 \sin(x)$$

$$f(-x) = -f(x)$$

odd